Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam D · Form 1: Multi-step reasoning

Multiple Choice Questions: Select the best answer. (1 point each)

1. Blocks m and 3m touch on a frictionless floor. Push inward on the small block or inward from the opposite side on the large block, using equal force F. By what factor is contact larger when pushing the small block?

(A) 4

(B) 3

(C) 1/3

(D) 1

2. A hanging m1 = 5 kg is connected by an ideal rope to m2 = 2 kg on a 35° incline. An extra vertically downward force of 0 N acts on m2. m2 slides uphill with $\mu_k = 0.1$. Find acceleration with uphill positive for m2. Use g = 9.8 m/s².

(A) 2.582 m/s²

(B) 5.165 m/s²

(C) -5.165 m/s²

(D) 10.329 m/s²

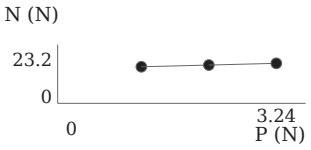
3. Press vertically down on a 2 kg block on a level floor. Use g=10 m/s². P (N) -> N (N): (1.24, 21.24); (2.24, 22.24); (3.24, 23.24). A logger displays twice each true horizontal reading. What slope appears on the displayed axes?

(A) 0.5

(B) 0.75

(C) 0.25

(D) 1



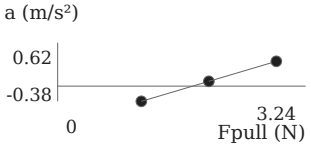
4. A 2 kg cart slides right against constant 2 N friction to the left. Double cart mass while holding external resistance at 2 N. Under the revised model, what is Y at X=2.24?

(A) 0.375

(B) 0.06

(C) 0.165

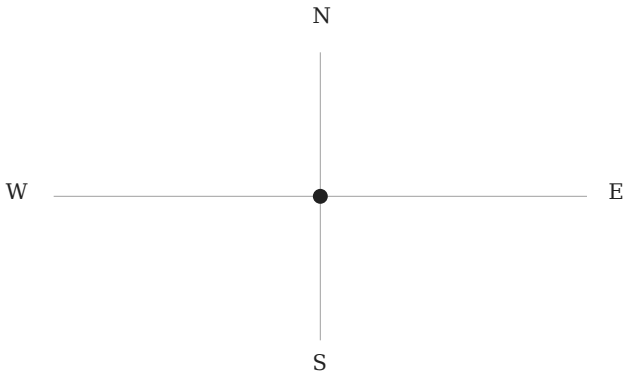
(D) -0.15



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Two students push a cart of mass 14 kg on a frictionless horizontal floor. A exerts 300 N at 25° north of east. B exerts 410 N at 39° south of west.

a. Draw and label the actual forces acting in the plane of the floor on the dot. Draw whole vectors, not components. **(1 pt)**



b. Show net-force x and y components, then find acceleration magnitude and direction using compass words. **(3 pt)**

6. Touching 2 kg and 3 kg carts accelerate on a frictionless floor when a hand pushes the 2 kg cart right. Fhand (N) -> Fcontact (N): (1.24, 0.744); (2.24, 1.344); (3.24, 1.944).

a. Build the model for two carts in contact from physics and predict both slope and intercept. **(2 pt)**

b. Push inward on the 3 kg cart to drive the 2 kg cart. Compare force magnitudes. Predict the new slope and intercept and explain physically what changed. **(2 pt)**

7. A mass-m cart slides right, pushed by F at θ below horizontal to produce acceleration a. The coefficient μ_k is known. Derive the required F with zero vertical acceleration and state the denominator condition. **(3 pt)**